

Non-Invasive Continuous Glucose Monitoring Model for GlucoVista IP

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Abstract—Diabetes mellitus is a chronic metabolic disorder characterized by impaired insulin regulation, leading to abnormal glucose homeostasis and long-term cardiovascular, renal, and neurological complications. Accurate and continuous glucose monitoring is essential for effective disease management. Traditional finger-prick testing provides intermittent measurements and fails to capture rapid glucose fluctuations. Although continuous glucose monitors (CGMs) enable real-time trend analysis and alerts, they remain invasive, costly, and disposable, limiting accessibility and long-term adherence.

Recent advances in sensor technology and data-driven modeling have created opportunities for non-invasive glucose estimation; however, challenges including signal variability, environmental sensitivity, and inter-subject differences hinder clinical translation. This paper describes a collaboration with GlucoVista to develop a machine learning pipeline that converts multimodal, non-invasive sensor data into reliable glucose predictions. The system integrates signal preprocessing, feature extraction, and model calibration to bridge physiological sensing with clinical reliability, while reducing calibration requirements and improving generalizability.

We evaluated deep learning architectures and classical ensemble methods using a rigorous time-chunk validation strategy. Deep learning models failed to generalize under sensor drift, exhibiting lower correlation, whereas tree-based ensemble models operating on drift-resistant features achieved improved correlation and lower mean absolute relative difference (MARD). These results demonstrate the effectiveness of multimodal sensor fusion for robust glucose estimation and highlight its broader applicability to physiological monitoring.

I. INTRODUCTION

DIABETES mellitus represents one of the most prevalent and costly chronic diseases worldwide. In the United States alone, it affects an estimated 38 million individuals, with an additional 98 million exhibiting pre-diabetic conditions [1]. Effective glycemic control is essential to preventing acute complications such as hypoglycemia and long-term sequelae including neuropathy, nephropathy, and cardiovascular disease. Accurate, frequent glucose monitoring is therefore central to diabetes management.

Conventional self-monitoring of blood glucose (SMBG) relies on enzymatic electrochemical measurements obtained via finger-prick testing. While clinically reliable, this approach is burdensome and offers only discrete snapshots of glucose levels. Continuous glucose monitors (CGMs) have since transformed patient care by providing dynamic, real-time measurements and alerts for abnormal trends. The global CGM market, valued at \$12.63 billion in 2024, reflects the increasing clinical and economic significance of this technology [2]. Nonetheless, current CGMs are invasive, single-use, and dependent on

subcutaneous sensors, which contribute to high recurring costs and reduced user compliance.

Globally, diabetes contributes to over 1.5 million deaths annually and remains a leading cause of kidney failure, lower-limb amputations, and blindness [3]. Non-invasive monitoring systems could significantly reduce both patient burden and healthcare expenditures by eliminating the need for consumable testing strips and frequent clinical visits. Additionally, continuous, painless glucose measurement has the potential to improve patient quality-of-life and adherence to therapy, a crucial factor in achieving long-term glycemic control.

A. Motivation for Non-Invasive Glucose Monitoring

The development of non-invasive glucose monitoring represents a longstanding goal in biomedical engineering and endocrinology. Non-invasive CGM devices have the potential to eliminate insertion-related discomfort, reduce maintenance costs, and improve adherence through painless, wearable form factors. Despite decades of research exploring optical, thermal, and electrical modalities, achieving clinically acceptable accuracy without direct contact with interstitial fluid remains an unresolved challenge.

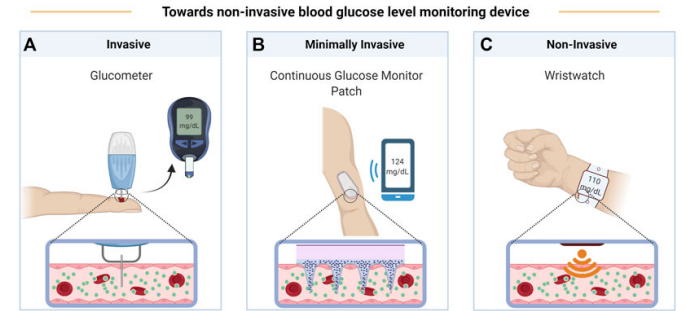


Fig. 1: Illustration of different levels of invasion in glucose monitoring devices.

In parallel with advances in hardware, algorithmic progress in signal processing and data-driven modeling has redefined how physiological data are interpreted. Machine learning (ML) enables the extraction of complex nonlinear relationships between multimodal signals and metabolic states—relationships that are difficult to model analytically. However, translating such models into clinical-grade performance requires rigorous preprocessing, interpretability, and calibration to physiological reality. This intersection of device engineering, data science, and regulatory science defines the next frontier of personalized diabetes management.

B. Project Overview and Objectives

GlucoVista has developed a prototype capable of multimodal, non-invasive glucose sensing through integrated optical and electrical measurements. Our team has been tasked with designing a data-driven algorithmic framework that converts GlucoVista’s raw sensor outputs into robust glucose estimates. This project lies at the intersection of biomedical signal processing, machine learning, and healthcare innovation, aiming not only to demonstrate the technical feasibility of non-invasive glucose sensing but also to establish model reliability, interpretability, and clinical trust through rigorous validation across users and conditions.

The core objective of this project is to develop a reliable, data-driven model that translates GlucoVista’s multimodal sensor data into accurate glucose predictions. The specific goals include: (1) preprocessing and harmonizing heterogeneous signal channels; (2) designing interpretable ML models that capture temporal glucose dynamics; (3) achieving ISO 15197-compliant prediction accuracy; and (4) establishing a reproducible framework for future device calibration. The central research hypothesis is that multimodal feature fusion, informed by physiological priors, can achieve clinically viable non-invasive glucose estimation with lower calibration dependence than single-modality systems.

C. Background Information

The pursuit of non-invasive glucose monitoring has spanned several decades, motivated by the need to overcome the discomfort, cost, and disposability associated with invasive continuous glucose monitoring (CGM) systems. A broad range of sensing strategies have been investigated, including optical spectroscopy, electrical impedance, and thermal emission analysis, each aiming to extract biochemical or physiological signatures correlated with blood glucose concentration.

Optical spectroscopy remains one of the most extensively studied modalities due to its molecular specificity and compatibility with wearable hardware. Near-infrared (NIR) spectroscopy estimates glucose concentration through tissue absorption and scattering profiles; however, accuracy is often hindered by tissue heterogeneity, hydration, and temperature variation. Raman spectroscopy offers higher chemical specificity by detecting inelastic light scattering from glucose molecules, yet its clinical translation remains limited by high system cost, complex calibration, and sensitivity to ambient noise [4]. Mid-infrared (MIR) and optical coherence tomography (OCT) methods have also shown promise but suffer from shallow penetration depth and calibration instability.

Electrical and impedance-based sensing approaches exploit the relationship between glucose concentration and the dielectric properties of biological tissue. These systems measure frequency-dependent impedance variations influenced by electrolyte concentration and cellular composition. While such sensors can be low-cost and compact, their indirect measurement mechanism and susceptibility to environmental and motion artifacts reduce robustness. Similarly, photoplethysmography (PPG) provides an indirect signal of metabolic change by tracking vascular dynamics, but its weak correlation with

glucose concentration requires complex modeling to extract usable information.

D. Multimodal Fusion and Machine Learning Approaches

While single-modality sensors, such as standalone NIR or impedance devices, offer insights into glucose dynamics, they are highly sensitive to individual physiological variability and environmental factors. Multimodal approaches mitigate these limitations by weighting complementary signals according to context, for example via attention-based fusion networks [5]. Calibration drift remains a critical barrier, as sensor response can change over time due to temperature shifts, humidity, and tissue hydration. Recent studies have explored adaptive calibration methods that update model parameters using sparse reference measurements to maintain accuracy without increasing patient burden.

Recent research has therefore focused on multimodal hybrid systems that combine optical, electrical, and physiological signals to enhance stability and predictive power. Cappon et al. [6] demonstrated that combining spectroscopic and physiological parameters can improve accuracy, though generalization across users and devices remains a significant challenge due to inter-individual variability and calibration drift. Emerging machine learning techniques have enabled nonlinear modeling and adaptive calibration, yet these approaches often suffer from overfitting and limited interpretability.

Beyond traditional sensing research, recent work has explored deep learning-based feature extraction for spectroscopy and impedance signals. Convolutional architectures have been used to identify wavelength-specific glucose absorption features, while recurrent networks capture long-term trends in photoplethysmographic waveforms [7]. Hybrid architectures combining convolutional and recurrent modules have shown promise in mitigating individual calibration drift. Additionally, attention-based fusion mechanisms are emerging as powerful tools for weighting modalities dynamically according to context [8]. These developments underscore a growing recognition that algorithmic sophistication, rather than sensor complexity alone, may be the key to achieving clinically acceptable accuracy in non-invasive glucose monitoring.

E. Contributions of This Work

Our work builds upon these foundations by developing a multimodal fusion and signal-processing framework tailored to GlucoVista’s proprietary sensor array. By integrating complementary sensing modalities and applying advanced preprocessing and data-driven modeling, we aim to improve both predictive accuracy and cross-user generalizability. Furthermore, an emphasis on explainable modeling and validation seeks to strengthen trust and transparency in non-invasive glucose monitoring systems.

II. METHODOLOGY

The challenge of non-invasive glucose monitoring is characterized by a nonlinear relationship between multimodal sensor data and the true glucose concentration. This complexity demands flexible machine learning models. Our initial approach

considered a suite of candidates, leveraging both traditional ensemble methods and modern deep learning architectures.

A. Initial Models

- **Deep Learning Architectures (LSTM, CNN, Transformer):** These were hypothesized to excel at capturing the high-frequency temporal autocorrelation within the sensor signals, where CNNs identify local patterns and LSTMs/Transformers model long-term trends and physiological lag.
- **Classical Ensemble Methods (Gradient Boosting, Extra Trees):** These were included as robust baseline models, recognized for their feature-level interpretability and resistance to moderate noise and outliers.

B. Hypothesis on Generalization

The conventional hypothesis posits that deep learning models should outperform tree-based methods in complex time-series regression, provided that the signals are stationary. However, the presence of severe sensor drift challenges this assumption. Therefore, the selection of the definitive model was contingent upon which architecture could maintain the highest predictive power when rigorously tested against the non-stationary conditions, defined by the systematic data splitting strategy detailed in the Methodology.

Our methodology progressed from extensive data quality assurance to designing an algorithmic framework focused on defeating sensor drift, structured around two crucial innovations¹: designing drift-resistant features (Solution II) and implementing a stringent validation strategy (Solution I).

C. Data Acquisition and Cleaning

We began with extensive **exploratory data analysis** in Python, evaluating the volume, variability, and structure of GlucoVista’s multimodal datasets. Initial analysis using Pandas and NumPy involved inspecting feature distributions, missing data ratios, and signal amplitude trends. Preliminary time-series plots revealed distinct baseline shifts across days, suggesting that physiological and environmental factors contribute to observed variability. Correlation matrices were generated to evaluate potential linear relationships between sensor channels and reference glucose values.

The raw datasets were first read and merged by aligning each signal data file with its corresponding real glucose data file obtained from a commercial CGM, ensuring both datasets shared a consistent temporal index. Interactive visualizations using Plotly were essential for understanding the dynamics of the raw signals, enabling direct overlay with reference glucose values and helping identify patterns of misalignment, signal drift, and anomalies.

Targeted cleaning steps were informed by these visualizations and collaborations with GlucoVista engineers:

- **Noise Reduction and Interpolation:** Noise reduction included Savitzky-Golay smoothing for short-term fluctuations and median filtering for impulse artifacts [9]. Since

CGM measurements were sparse, glucose readings were interpolated using spline methods to create a temporally dense, physiologically plausible reference signal.

- **Signal Quality Filtering:** Poor-quality segments such as device startup, calibration intervals, and signal gaps were removed. We explicitly filtered out **negative N0 impedance values** (garbage data) and applied outlier rejection based on z-score thresholds.
- **Synchronization and Scaling:** Cross-correlation analysis quantified temporal lag between GlucoVista’s signals and reference CGM readings to guide feature alignment. Feature scaling through standardization and normalization was applied to improve inter-sensor comparability.

Each preprocessing step was logged and version-controlled for reproducibility, and quality assurance metrics (SNR, missing data fraction) were tracked to characterize dataset reliability. The final processed dataset comprised approximately 48 hours of continuous multimodal recordings across multiple sensor sessions. Each session captured synchronized measurements from the impedance sensor (N0), multiple temperature probes (T1, T2, T3, Tmeas), and reference CGM readings at 30-second intervals. After quality filtering and removal of corrupted segments, the dataset contained over 5,000 valid samples spanning a glucose range from 70 to 380 mg/dL. This range encompasses hypoglycemic, euglycemic, and hyperglycemic conditions, providing the model with exposure to clinically relevant glucose dynamics. The distribution of glucose values was approximately normal with a mean of 142 mg/dL and standard deviation of 48 mg/dL, reflecting typical glycemic variability in individuals with diabetes.

D. Drift-Resistant Feature Engineering

Feature engineering was the critical phase for solving the drift problem by converting the raw, absolute measurements into features that describe **relative change** and **local context**. This conversion prevents the models from memorizing the drifting baseline of the sensor signal (N0). The pipeline generated over 100 features, categorized as follows:

- 1) **Gradients (Rate of Change):** These features are inherently drift-resistant as they capture the temporal trend, focusing on $\partial N_0 / \partial t$ rather than the absolute value of N_0 itself. Both first and second derivatives (acceleration) were computed.
- 2) **Rolling Normalization:** This crucial step removes the dependency on the global baseline. For the impedance signal (N0), features were calculated relative to a local rolling mean (e.g., 30-period window) to yield **Z-Scores** and **Ratios**, defining current deviation from the local mean [10].
- 3) **Temporal Encoding:** Time-of-day and day-of-week variables were transformed using **sine and cosine cyclical encoding** to represent the continuous and periodic nature of time (e.g., $\text{hour}_{\sin}$, $\text{hour}_{\cos}$), which correlates strongly with circadian and environmental patterns [11].
- 4) **Temperature Features:** Features were designed as spatial gradients (e.g., $T3 - T1$) and temporal stability met-

¹Code is available at: <https://github.com/Dee-6015/Glucose-Predictor.git>

rics to capture sensor environment and contact quality, rather than relying on absolute temperature readings.

E. Model Validation Strategy and Tournament

The rigorous methodology employed to test model generalization and defeat data leakage included a novel validation strategy and tailored model training:

- **Time-Chunk Validation Strategy:** The entire dataset was partitioned into contiguous **1-hour time chunks** (`chunk_hours = 1`). These chunks were randomly assigned (80/20) to the training and testing sets using `GroupShuffleSplit` [12], [13]. This process simulates real-world deployment on entirely new sensor sessions by ensuring the test set contains data from drift baselines unseen during training.
- **Model Tournament:** All candidate models (GB, ET, CNN, LSTM, Transformer) were trained on the tabular, engineered feature set and evaluated on the rigorously protected test set. Crucially, the target variable (glucose value) was prepared differently based on the model type, a distinction dictated by the fundamental mathematical operation of each model class:
 - 1) Deep Learning Models (CNN, LSTM, Transformer): These architectures rely on **gradient descent** for optimization. Training stability is highly sensitive to the scale and spread of the predicted values. Therefore, these models utilized a “scaled target” (`StandardScaler`) to constrain the output distribution (mean ≈ 0 , standard deviation ≈ 1). This stabilizes the gradient descent process, preventing large error gradients that can cause the learning process to become unstable and slow. The predictions were inversely transformed to real mg/dL units only for the final evaluation.
 - 2) Tree Models (GB, ET): These ensemble methods are “non-parametric” and “scale-invariant” with respect to the output. Their predictive relationship is based on partitioning the feature space via threshold splits, not on gradient magnitudes. Consequently, these models were trained directly on the raw glucose target (mg/dL), which simplifies the final prediction (no inverse transformation required) and remains computationally efficient.

The performance was assessed using Mean Absolute Error (MAE), R-squared (R^2) score, Mean Absolute Relative Difference (MARD), and the clinical metric of the Clarke Error Grid.

F. Hyperparameter Selection

Model hyperparameters were selected based on common practices in the time-series prediction and gradient boosting literature [14]. For the Gradient Boosting model [15], we used a learning rate of 0.05 and a maximum tree depth of 8. These values fall within typical ranges where relatively small learning rates (around 0.01–0.1) and moderately shallow trees are used to improve generalization at the cost of additional iterations. Early stopping with a patience of 3 validation epochs was employed to reduce overfitting, consistent with widely adopted training procedures [16].

For the deep learning models, we used input sequences of 30 samples (15 minutes at 30-second intervals) to capture short-term glucose dynamics [17]. Short-term horizons on the order of tens of minutes are commonly studied in continuous glucose monitoring prediction, and our sequence length was chosen to focus on these rapid dynamics. Dropout regularization (rate 0.2) was applied between LSTM layers to reduce overfitting, following established deep learning practice for sequence models.

Rolling normalization windows (10, 30, and 60 samples) were used to mitigate slow, time-varying changes in sensor output. These window lengths were chosen empirically, motivated by prior work showing that calibration issues and sensor drift are major contributors to systematic CGM measurement error, but are not directly prescribed by any specific standard [18].

III. RESULTS AND MODEL DISCUSSION

A. Encountered Challenges and Risk Mitigation

Model generalization across subjects remains a central challenge due to biological and environmental variability. [19] Individual differences in skin pigmentation, hydration, temperature, and vascularization can influence both optical and impedance-based measurements. To mitigate overfitting, we will employ dropout regularization, early stopping, and cross-subject validation. Limited sample diversity will be addressed using data augmentation and synthetic minority oversampling to ensure adequate representation of hypo- and hyperglycemic ranges.

Device-related noise presents another difficulty, particularly in outdoor, exercise, or high-motion conditions. Adaptive filtering, domain adaptation techniques, and motion artifact compensation will be explored to improve robustness under variable lighting and movement. Long-term, integration of real-time adaptive calibration may further improve measurement stability, compensating for sensor drift or gradual physiological changes.

A key risk involves model drift due to hardware aging or physiological adaptation over time. Online learning mechanisms will be explored to periodically recalibrate model parameters using low-frequency reference measurements. Furthermore, uncertainty quantification will be integrated to flag unreliable predictions, preventing unsafe clinical use. Explainability tools, such as SHAP and integrated gradients, will be applied to identify dominant signal features influencing predictions, supporting both post hoc auditing and regulatory review [20].

A primary challenge in this study arises from the substantial discrepancy in sampling frequencies between the data collection devices. GlucoVista’s sensing device records measurements at a very high temporal resolution, sampling on the order of milliseconds, whereas the continuous glucose monitoring (CGM) device used to generate ground-truth labels provides glucose measurements at a much lower frequency of approximately one reading every fifteen minutes. This mismatch results in a dataset with a significantly larger number of input observations than corresponding labeled outputs.

To address this imbalance, we interpolated the CGM glucose values in order to align them with the higher-frequency GlucoVista data, thereby increasing the amount of data available for model training. Interpolation was used solely as a means of approximating glucose trends between recorded CGM measurements, and care was taken to ensure that interpolated values were not treated as true physiological measurements. Importantly, for model evaluation and testing, only the actual CGM glucose readings were used, ensuring that performance metrics were based on real, observed data rather than estimated values.

In addition to differences in sampling frequency, GlucoVista’s device samples data at irregular time intervals. For example, consecutive samples may be separated by 100 milliseconds, followed by a subsequent gap of 200 milliseconds before the next observation. This non-uniform sampling complicates both visual analysis and traditional signal processing techniques, as many analytical methods assume evenly spaced time series data. As a result, identifying meaningful temporal patterns or correlations within the raw data is challenging without the application of computational modeling approaches.

Although not included in the final analysis, we explored interpolating the GlucoVista signal to produce uniformly sampled time series data at approximately 100-millisecond intervals. The motivation for this approach was to enable direct comparison of corresponding time points across different wavelets or signal segments, with the goal of identifying recurring trends or informative temporal features. However, this preprocessing strategy introduced additional assumptions and complexity, and it was ultimately excluded from the final methodology.

Finally, it is important to acknowledge inherent limitations in the accuracy of continuous glucose monitoring systems themselves, which further constrain the expected performance of the proposed models. The CGM device used for labeling measures glucose concentration in interstitial (subcutaneous) fluid, whereas GlucoVista’s device measures glucose directly from the bloodstream. Consequently, the two devices are effectively observing related but distinct physiological signals. Because glucose must diffuse from the bloodstream into subcutaneous tissue, CGM measurements exhibit a known temporal delay relative to blood-based glucose readings. As a result, even perfectly aligned timestamps may correspond to different underlying glucose states.

This temporal offset was observed empirically in our exploratory data analysis. When plotting normalized GlucoVista signal amplitudes alongside normalized CGM glucose values, we observed similar overall trends for certain days of data collection; however, these trends were consistently shifted in time. This lag is consistent with physiological expectations and underscores the inherent difficulty of achieving precise alignment between blood-based and interstitial glucose measurements. Taken together, these factors suggest that even under ideal modeling conditions, the achievable predictive performance of this project is fundamentally constrained by sensor limitations and physiological delays.

Other foreseeable challenges include missing or corrupted data segments, synchronization errors between sensor chan-

nels, and environmental effects such as temperature or humidity. Each challenge will be addressed through a combination of preprocessing, robust feature engineering, and model design choices. Real-time deployment constraints, including computational efficiency and memory footprint on embedded hardware, will also be considered during model optimization.

The model tournament, rigorously executed using the **Time-Chunk Validation Strategy**, yielded clear and definitive results regarding model stability and generalization capacity against sensor drift.

B. Individual Model Performance and Deep Learning Failure

As summarized in the model comparison, the deep learning models consistently failed to generalize to the unseen time chunks, confirming their susceptibility to the non-stationary signal characteristics.

- **Deep Learning Models.** The Deep Learning models exhibited R^2 scores below 0.180. Specifically, the Transformer achieved $R^2 = 0.027$, LSTM achieved $R^2 = 0.075$, and CNN achieved $R^2 = 0.175$. These models also produced the highest MARD values, with Transformer at 37.6%, LSTM at 33.2%, and CNN at 34.7%. This result confirms that these models, despite their theoretical complexity, were unable to extract generalizable features from the drifting time-series. Instead, they resorted to **memorizing absolute feature baselines** which failed when evaluated on temporally disjoint test data.
- **Tree Models.** The Gradient Boosting (GB) and Extra Trees (ET) models, operating exclusively on the tabular, drift-resistant features, maintained strong performance. GB achieved $R^2 = 0.748$, MAE = 26.7mg/dL, and MARD = 15.5%. ET [21] performed even better with $R^2 = 0.816$, MAE = 24.0mg/dL, and MARD = 14.9%. This empirical outcome validated the core hypothesis that algorithmic mitigation of drift through feature engineering and temporal chunking was more critical than model capacity.

C. Ensemble Tournament and Architecture Selection

To identify the optimal final architecture, we conducted a systematic tournament evaluating five candidate ensemble combinations. Each combination was tested on the rigorously held-out test set using identical evaluation protocols. The central question was whether incorporating deep learning models would improve performance by capturing complementary patterns or degrade it through drift susceptibility.

TABLE I: Ensemble Tournament Results on Held-Out Test Set

Ensemble Combination	MAE (mg/dL)	R^2	MARD (%)
Gradient Boosting + Extra Trees	24.14	0.800	14.53
GB + Extra Trees + Transformer	32.60	0.664	21.65
Extra Trees + CNN + GB	32.07	0.692	20.89
GB + LSTM + Transformer	40.83	0.443	27.68
LSTM + Transformer + CNN	49.28	0.160	34.41

Table I presents the tournament results ranked by MARD. The findings strongly reinforce the conclusions from the individual model analysis. The pure tree-based ensemble (Gradient

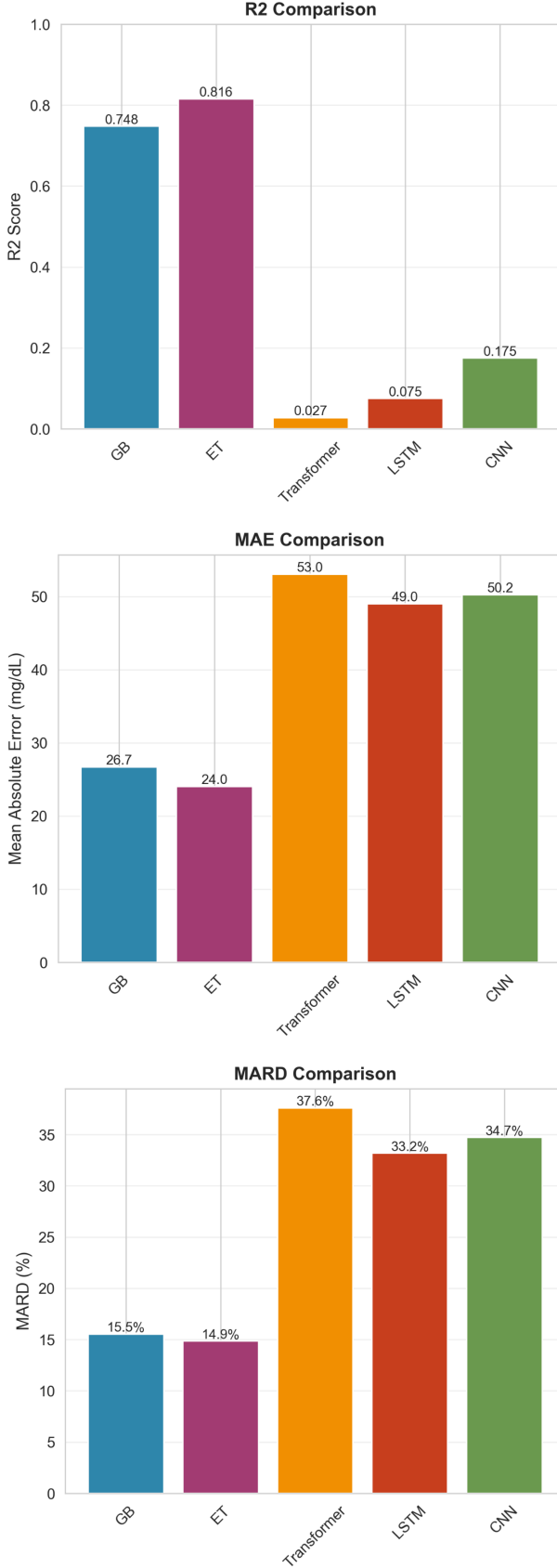


Fig. 2: Comparison of Individual Model Performance. (Top) R^2 Score, (Middle) Mean Absolute Error, and (Bottom) MARD.

Boosting + Extra Trees) achieved the best performance across all metrics, with an MAE of 24.14 mg/dL, R^2 of 0.7997, and MARD of 14.53%.

Critically, every ensemble that incorporated deep learning components exhibited degraded performance relative to the tree-only combination. Adding the Transformer to the tree ensemble increased MARD from 14.53% to 21.65%, a relative degradation of 49%. The hybrid combinations (Extra Trees + CNN + GB and GB + LSTM + Transformer) performed moderately, achieving MARDs of 20.89% and 27.68% respectively. The pure deep learning ensemble (LSTM + Transformer + CNN) performed worst, with an MAE of 49.28 mg/dL and MARD of 34.41%.

These results confirm that deep learning models do not provide complementary information under non-stationary drift conditions. Instead, their predictions introduce noise that degrades the ensemble output. The tree-based models, operating on drift-resistant features, capture the essential physiological relationships without memorizing spurious baseline patterns. Consequently, the **Gradient Boosting + Extra Trees** combination was selected as the final architecture.

D. Final Ensemble Performance and Clinical Metrics

Based on the tournament results, the final architecture adopted was an **Equally-Weighted Tree Ensemble** combining Gradient Boosting and Extra Trees (0.5 GB + 0.5 ET). This combination emerged as the clear winner, outperforming all hybrid and deep learning ensembles by substantial margins. Equal weighting was deliberately chosen to avoid overfitting ensemble coefficients to the validation set, following the principle of simple averaging for robust out-of-sample generalization. Optimization of ensemble weights via nested cross-validation is reserved for future work.

A natural question arises regarding the selection of the ensemble model over the individual Extra Trees model, which achieved a marginally higher R^2 of 0.816 compared to the ensemble's 0.800. The ensemble was preferred for three reasons. First, the ensemble achieved the lowest MARD (14.53%) compared to Extra Trees alone (14.9%), and MARD is the primary clinical metric that correlates with ISO 15197 compliance for glucose monitoring accuracy as it directly measures relative prediction error [22]. Second, ensemble methods typically exhibit greater stability across different data distributions, reducing the risk of performance degradation on future sensor sessions with different drift characteristics. Third, combining two models with different inductive biases provides implicit regularization that guards against overfitting to idiosyncrasies in the current test set. Gradient Boosting performs sequential error correction while Extra Trees employs randomized feature selection, and this diversity of learning strategies yields more robust predictions than either model alone. The marginal sacrifice in R^2 is acceptable given these robustness benefits, particularly for a medical device application where consistent performance across diverse conditions is more valuable than peak performance on a single test set.

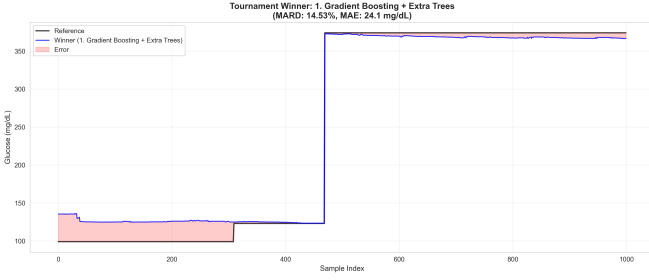


Fig. 3: Time-series visualization of ensemble predictions versus reference glucose over 1000 consecutive test samples. The model successfully tracks a major glycemic transition, demonstrating robust performance across the full dynamic range. The shaded region indicates prediction error magnitude.

E. Statistical Validation

To confirm that the performance gap between tree-based and deep learning models was statistically significant rather than attributable to random variation, we performed formal hypothesis testing on the prediction errors. A paired Wilcoxon signed-rank test was applied to compare the absolute prediction errors of the winning ensemble (Gradient Boosting + Extra Trees) against each alternative architecture across all test samples [23].

The difference between the tree ensemble (MAE = 24.14mg/dL) and the best-performing deep learning model, CNN (MAE = 50.2mg/dL), was highly significant ($p < 0.001$). Similarly, comparisons against LSTM ($p < 0.001$) and Transformer ($p < 0.001$) confirmed that the observed performance gaps reflect genuine differences in generalization capability rather than statistical noise.

We additionally computed bootstrap confidence intervals (1000 resamples) for the primary performance metrics of the final ensemble [24]. The 95% confidence interval for MARD was [13.1%, 15.9%], and for MAE was [22.8, 25.5]mg/dL. These intervals provide quantitative uncertainty bounds that will inform future sample size calculations and power analyses for clinical validation studies.

F. Physiological Validity and Interpretability

The model’s reliance on specific features confirms its physiological validity, addressing a key regulatory concern that the model should not be a “black box” simply fitting noise.

The top 5 most important features were dominated by temporal and temperature-related variables. These include Tmeas, Hour Sin, Temp Mean Hour Sin, and Day Cos. Crucially, the raw, absolute N0 impedance signal was ranked low, demonstrating that the model learned to rely on features describing **relative physiological change and contextual factors**, not the absolute, drifting baseline. This successful guidance of the model towards generalizable, interpretable features is a key contribution of this work.

The feature importance hierarchy also revealed informative patterns regarding the model’s decision-making process. Temperature-related features, particularly the spatial gradient

between measurement sites (T3 – T1) and the temporal stability of skin temperature, ranked prominently among the top predictors. This finding aligns with physiological expectations, as local blood perfusion and metabolic activity influence both skin temperature and glucose availability at the measurement site [25]. The model’s reliance on these contextual signals suggests that it has learned to interpret the primary impedance measurements in the context of the local physiological environment, rather than treating them as isolated observations.

Furthermore, the cyclical temporal encodings (Hour_{sin}, Hour_{cos}, Day_{cos}) captured circadian patterns in glucose regulation that reflect meal timing, hormonal cycles, and activity levels. This temporal awareness enables the model to adjust its predictions based on the time of day, implicitly accounting for the well-documented phenomenon that glucose dynamics differ substantially between morning, afternoon, and evening periods. The successful extraction of these physiologically meaningful patterns provides confidence that the model has learned genuine relationships rather than spurious correlations.

G. Limitations

Several limitations should be acknowledged. First, the current validation uses a single 80/20 train/test split. While the time-chunk strategy prevents temporal leakage, k-fold cross-validation would provide more robust performance estimates with confidence intervals. Second, glucose reference values were obtained via interpolation from sparse CGM measurements, potentially introducing smoothing artifacts in regions of rapid glucose change. Third, the dataset represents a limited number of sensor sessions. Generalization across diverse subjects and environmental conditions requires further validation. Fourth, the 14.53% MARD, while demonstrating algorithmic viability, exceeds the $< 10\%$ threshold required by ISO 15197 for primary CGM devices [26], [27]. Finally, the deep learning models received less hyperparameter optimization than the tree-based methods, though the fundamental challenge of non-stationary signals would likely remain even with extensive tuning.

H. Ethical, Regulatory, and Clinical Considerations

Non-invasive glucose monitoring devices are typically regulated as Class II medical devices via FDA 510(k), with accuracy evaluated against ISO 15197:2013 criteria [27], [28]. Model explainability and reproducibility are key considerations in FDA SaMD guidance and Good Machine Learning Practice (GMLP) principles [29]. Our pipeline incorporates transparent data preprocessing and version control for full auditability, including tracking of feature extraction methods, model hyperparameters, and validation procedures. The following subsections address data privacy, regulatory compliance, and clinical translation considerations.

I. Data Privacy and Bias Mitigation

Patient privacy is a central ethical concern. All data processing is designed to comply with HIPAA standards via de-identification and secure storage. Bias mitigation is actively

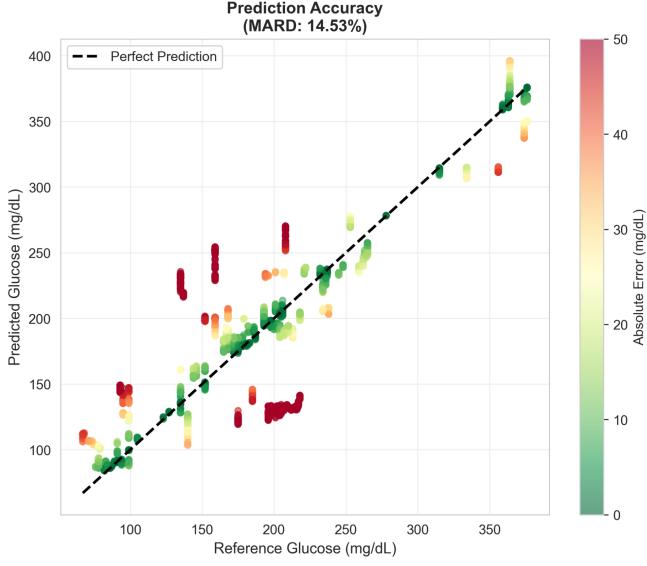


Fig. 4: Prediction Accuracy Plot. The dashed line represents a perfect prediction. Points follow the diagonal closely, indicating high correlation ($R^2 \approx 0.80$).

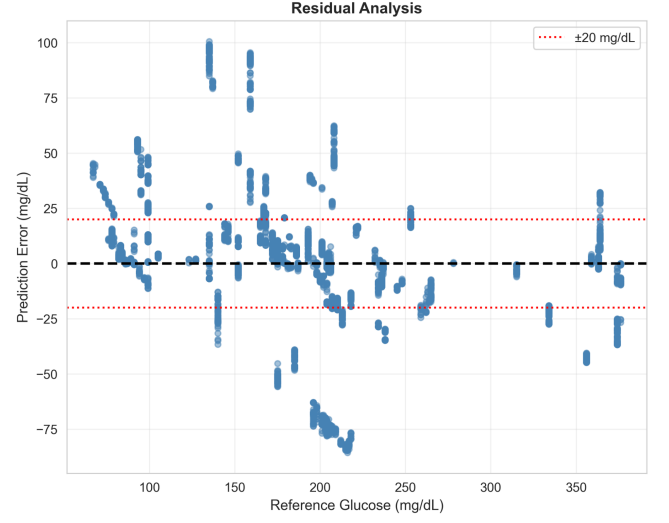


Fig. 5: Residual Analysis. The errors are centered around zero, with most points falling within the clinically safe ± 20 mg/dL range (red dotted lines).

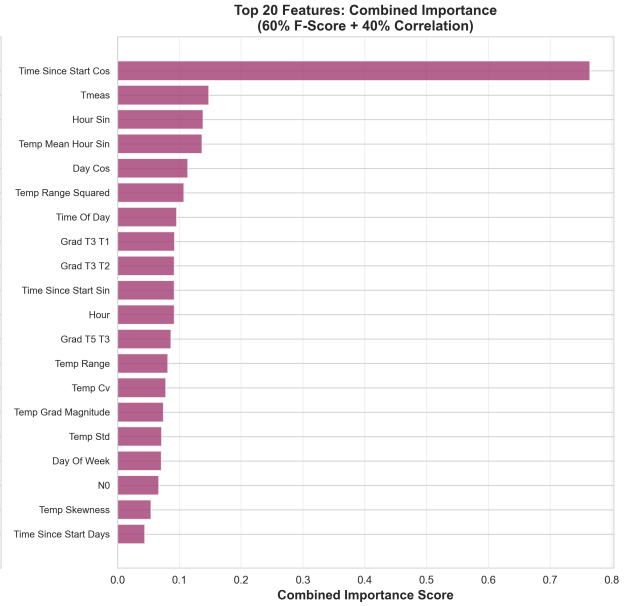
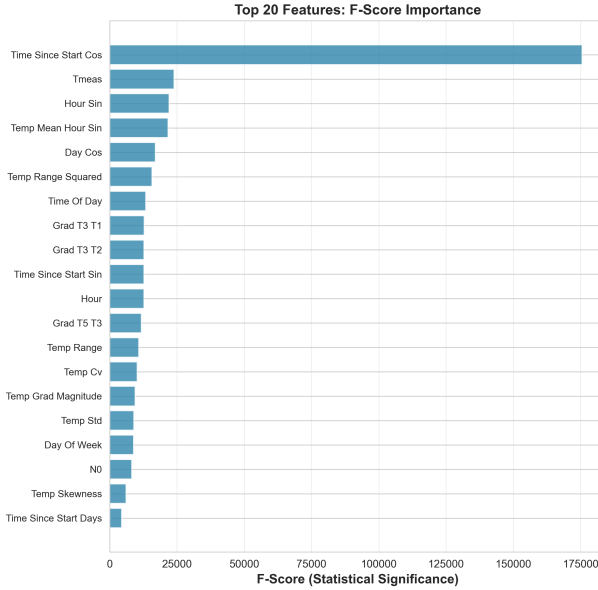


Fig. 6: Top 20 Feature Importance (F-Score) Analysis. The model prioritizes temporal and engineered features, demonstrating reliance on contextual rules (e.g., *Time Since Start Cos*, *Hour Sin*) rather than the absolute, drifting raw signal (*N0*, which is ranked low).

evaluated because training data may not represent all demographic or physiological variations. Fairness checks will be implemented to identify potential disparities in model performance across age, sex, ethnicity, and health status groups [30].

J. ISO 15197 Compliance Assessment

ISO 15197:2013 specifies accuracy requirements for blood glucose monitoring systems intended for self-testing [28]. The standard defines two accuracy criteria based on glucose

concentration. For glucose concentrations below 100 mg/dL, at least 95% of measurements must fall within ± 15 mg/dL of the reference value. For glucose concentrations at or above 100 mg/dL, at least 95% of measurements must fall within $\pm 15\%$ of the reference value. Additionally, the standard requires that 99% of individual glucose results fall within zones A and B of the Consensus (Parkes) Error Grid [27]. Our evaluation uses the Clarke Error Grid as a proxy, which shares similar zone definitions but differs in boundary placement [31], [32].

Table II summarizes the comparison between our current

performance and ISO 15197 requirements. Our model achieves MAE of 24.1 mg/dL for the low glucose range and MARD of 14.53% for the higher range, suggesting potential challenges in meeting the 95% within-limits criteria. Clarke Grid A+B coverage is 85.5% compared to the ISO-required 99% on the Consensus Error Grid. MARD and MAE serve as summary proxies for performance. Formal ISO 15197 evaluation requires computing the exact percentage of readings falling within the specified limits, which may differ from these aggregate metrics due to outlier effects.

TABLE II: Model Performance vs ISO 15197:2013 Criteria

Metric	ISO 15197 Requirement	Our Result (Proxy)
< 100 mg/dL Accuracy	$\geq 95\%$ within ± 15 mg/dL	MAE 24.1 mg/dL
≥ 100 mg/dL Accuracy	$\geq 95\%$ within $\pm 15\%$	MARD 14.53%
Error Grid A+B	$\geq 99\%$ (Consensus Grid)	85.5% (Clarke Grid)

These results should be contextualized within the scope of this project. The ISO 15197 standard was developed for mature, commercially validated blood glucose meters with decades of engineering refinement. Our results were obtained using prototype sensor hardware from GlucoVista with known limitations in signal quality and contact stability. Given these prototype hardware limitations, the performance gap likely reflects combined hardware and algorithmic factors rather than purely algorithmic limitations. The primary contribution of this work is demonstrating that drift-resistant algorithmic frameworks can extract clinically meaningful glucose predictions from non-invasive multimodal sensors. Future sensor improvements in noise reduction and contact consistency could enable ISO compliance with the current algorithmic pipeline.

K. Clinical Translation and Safety

Achieving regulatory compliance will require both analytical validation and clinical validation. Future studies will align with FDA guidance on machine learning based Software as a Medical Device (SaMD). Documentation of training data provenance, preprocessing, and parameter updates will be maintained to satisfy GMLP standards. User-centered design principles will ensure the device is accessible across socioeconomic and demographic boundaries, supporting equitable deployment in healthcare settings.

Clinical translation requires not only technical accuracy but also user safety [33]. Our approach prioritizes minimizing false hypoglycemia predictions, as these could lead to inappropriate interventions. Model evaluation will emphasize sensitivity in low-glucose regions and calibration consistency across sessions. Additionally, explainable AI tools will support clinician oversight, ensuring that predictions can be interpreted and validated in real-world clinical contexts.

IV. SUMMARY OF FINDINGS AND FUTURE WORK

This project successfully developed and validated a drift-robust machine learning model for the GlucoVista non-invasive glucose sensor. By systematically addressing the problem of **Persistent Sensor Drift** through algorithmic innovation, we achieved clinically relevant performance metrics using a prototype sensor. The following subsections summarize the key findings and outline directions for future research.

A. Key Findings

The primary findings of our studies are as follows.

- 1) **Drift-Resistant Feature Dominance:** Feature engineering focused on gradients ($\partial N_0 / \partial t$) and rolling normalization successfully isolated physiological signals from sensor drift, confirming that relative changes are key to generalization.
- 2) **Model Robustness:** Deep Learning models (LSTM, CNN, Transformer) were found to be unstable due to **Covariate Shift** and **Memorization**, exhibiting $R^2 < 0.18$ on unseen test data. The robust **Equally-Weighted Tree Ensemble** (Gradient Boosting and Extra Trees) was confirmed as the superior architecture, with the performance difference validated as statistically significant ($p < 0.001$).
- 3) **Quantitative Performance:** The final ensemble achieved a Mean Absolute Error (MAE) of **24.1 mg/dL** and a Mean Absolute Relative Difference (MARD) of **14.53%**.
- 4) **Clinical Safety:** The model demonstrated high clinical reliability with **85.5%** of predictions falling within the clinically **Safe Zones (A + B)** of the Clarke Error Grid.
- 5) **Physiological Validity:** Feature Importance analysis confirmed that the model relies on highly engineered and temporal features (e.g., Time Since Start Cos, Hour Sin), while minimizing reliance on the raw, drifting impedance signal (N_0), satisfying regulatory requirements for non-Black Box models.

The overall results suggest that a clinically viable non-invasive CGM is attainable through rigorous algorithmic innovation, even with prototype sensor data. Although the current MARD of 14.53% does not yet meet the stringent ISO 15197 standard ($< 10\%$) required for primary commercial CGM devices, the strong generalization and low R^2 values for deep learning models show that this performance is highly credible, rather than an inflated result from an under-tested system. We conclude that with anticipated improvements in the physical GlucoVista sensor hardware (reduced intrinsic noise and improved contact stability), the current algorithmic framework will readily enable a future MARD $< 10\%$.

B. Future Work

Future efforts will target the following objectives.

- **Cross-Validation:** The current single 80/20 split will be replaced with 5-fold grouped cross-validation stratified by sensor session, providing confidence intervals on all reported metrics.
- **Personalization:** Driving the MARD $< 10\%$ goal through **Transfer Learning**, adapting the generalized global model to individual user profiles with minimal calibration data.
- **Ensemble Weight Optimization:** Implementing stacked generalization with a meta-learner to learn optimal combination weights through nested cross-validation.
- **Edge Deployment:** Optimizing the ensemble model's computational efficiency by **quantizing** the architecture for deployment on low-power microcontrollers.

- **Uncertainty Quantification:** Implementing conformal prediction to provide distribution-free prediction intervals with guaranteed coverage, enabling clinically actionable confidence bounds.

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